Servicenow Administrator - NM Engg ServiceNow System

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

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📓 Automating Data Import and Relationship Mapping

Using Import Sets & Dot Walking

🔹 Introduction

Automating data import and relationship mapping is an essential process in modern IT systems (like ServiceNow) to efficiently transfer data from external sources into internal databases. This process ensures that data is not only imported but also properly linked with related records using reference fields.

🔹 Import Sets (Data Staging Concept)

An **Import Set** is a temporary table used to store incoming data before inserting it into the actual system tables.

👉 Purpose:

- Acts as a buffer between external data and system tables
- Prevents direct data corruption
- Allows validation and transformation

👉 Example:

Data from an Excel file containing employee details is first loaded into an import set table.

🔹 Transform Maps (Data Conversion Layer)

A **Transform Map** defines how the data from the import set is transferred to the target table.

👉 Key Functions:

- Field mapping (source → target)
- Data transformation
- Script execution for automation

👉 Example:

- emp_name → name
- dept → department

🔷 Automation in Data Import

Automation reduces manual effort by using:

✅ **Scripts in Transform Maps:**

- **onBefore**: Validate data
- **onAfter**: Modify target record
- **onComplete**: Final actions

✅ **Scheduled Imports:**

- Automatically fetch data from external sources like FTP or APIs

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🔷 Relationship Mapping

In databases, tables are connected using **reference fields**.

👉 **Example:**

- Employee → Department
- Incident → Caller

Instead of storing plain text, the system stores a unique ID (**sys_id**) to maintain relationships.

🔷 Dot Walking (Key Concept)

Dot Walking is a technique used to access fields from related tables using reference fields.

👉 **Syntax:**

reference_field.field_name

👉 **Example:**

If:

- caller_id is a reference to User table
- User table has email

Then:

caller_id.email

✓ This retrieves the email of the caller from the related table.

🔷 Relationship Mapping Using Dot Walking

During import, data may contain values like department names instead of IDs.

👉 Solution:

- Use scripts to:
    1. Search for the related record
    2. Link it using sys_id
    3. Create record if not found

👉 Example Script:

```
var dept = new GlideRecord('cmn_department');
dept.addQuery('name', source.department);
dept.query();

if (dept.next()) {
   target.department = dept.sys_id;
}
```

🔷 Real-Time Example

Scenario:

Import student data:

Name	Course
Kavin	CSE
Arun	ECE

Process:

1. Load data into Import Set
2. Create Transform Map
3. Use script:
    - Check if course exists
    - Create if not
    - Map student → course

◈ Advantages

✓ Automates large data imports
✓ Maintains data relationships
✓ Reduces human errors
✓ Improves data consistency
✓ Saves time and effort

Create records in the Sheet:

Description

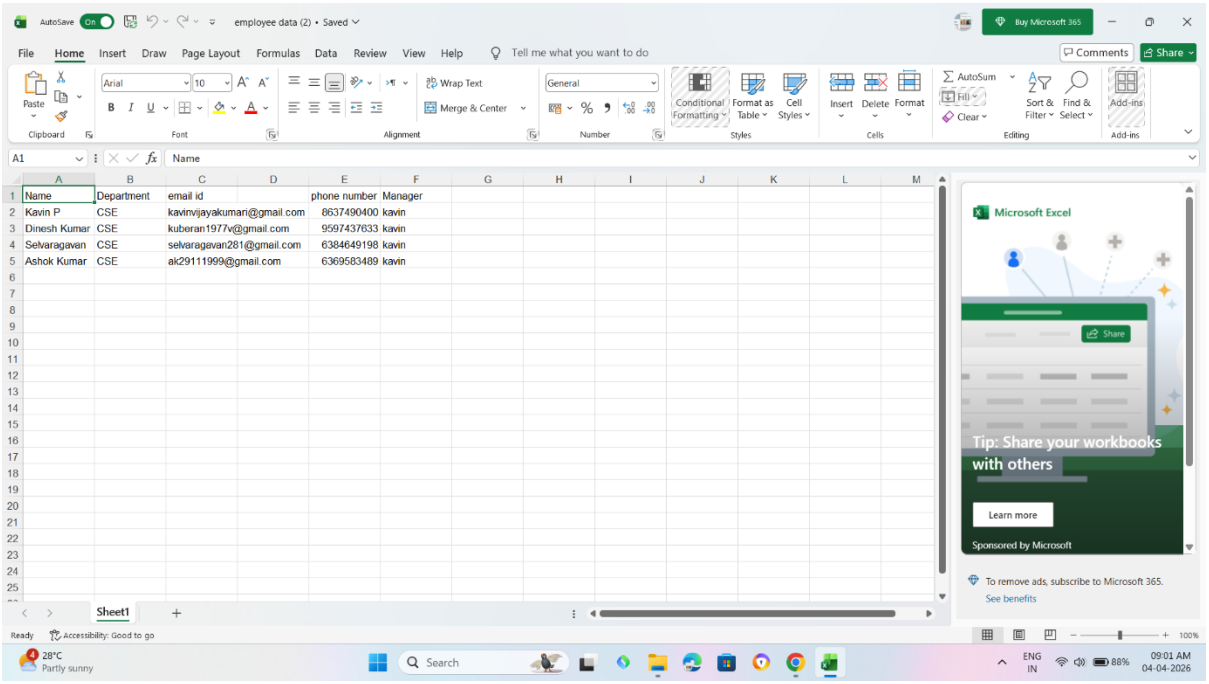
Open google sheet

Name the fields as name,email,department, manager and phone number

Enter some records

Save the sheet as: employee data

Download the file as .xlsx , .csv (or) .xml file



Creating Import Table(Staging Table):

Description

Open service now.

Click on All >> search for import sets

Select load data under system import sets

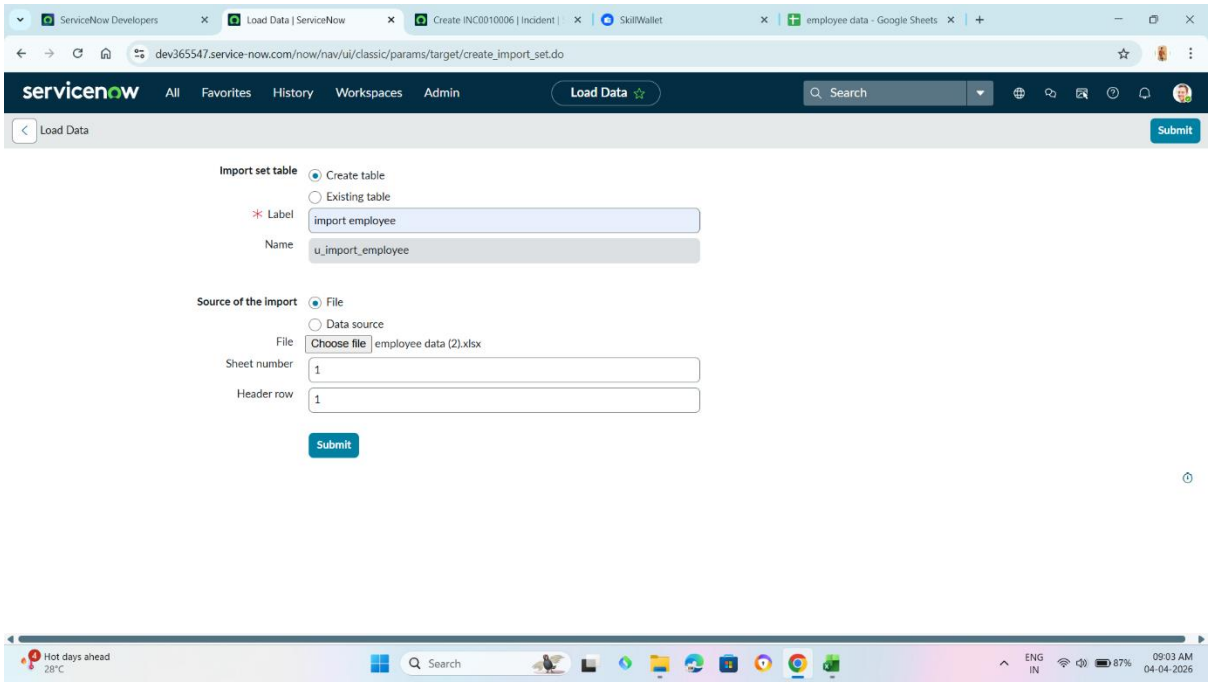
select create table

Give the the import set table name as: import employee

Choose the file which we created in the XL sheet.

7. The data is imported,click on personalized list

8. Add fields to the form.



Create transform map:

Open service now.

Click on All >> search for import set

Select create transform map under import sets

Click on new

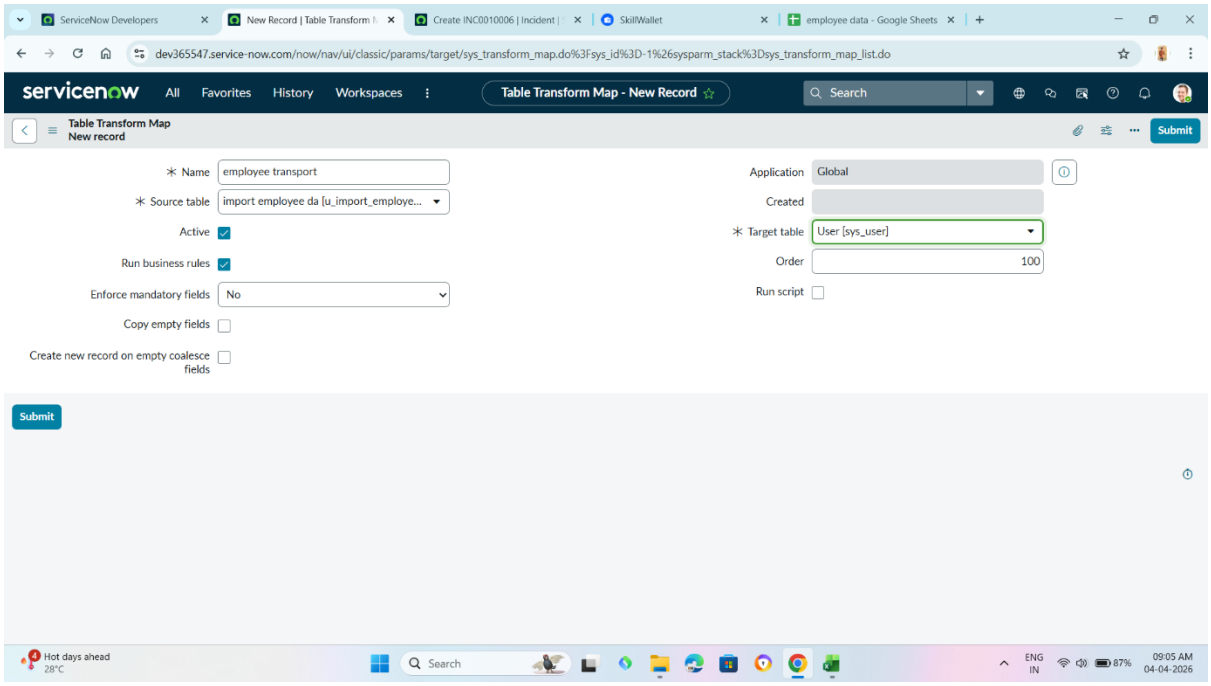
Fill the following details to create a new transform map

Give name as: Test

Select source table: @123

Select target table: user[sys_user]

Save the form.

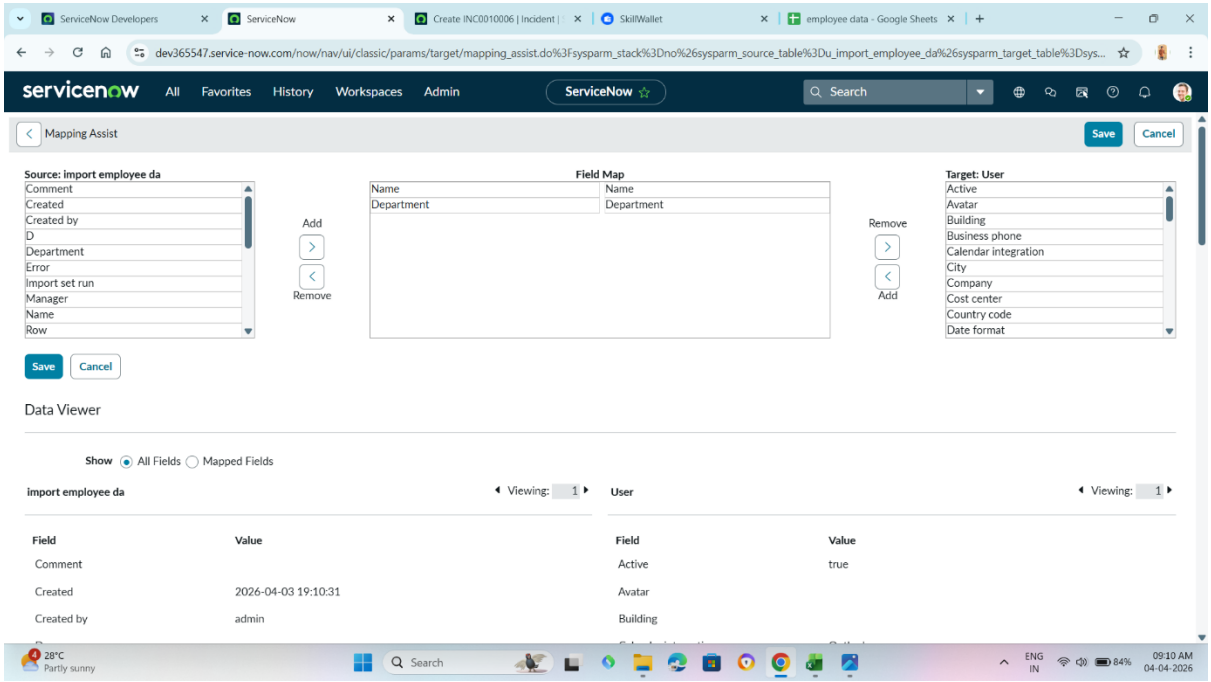


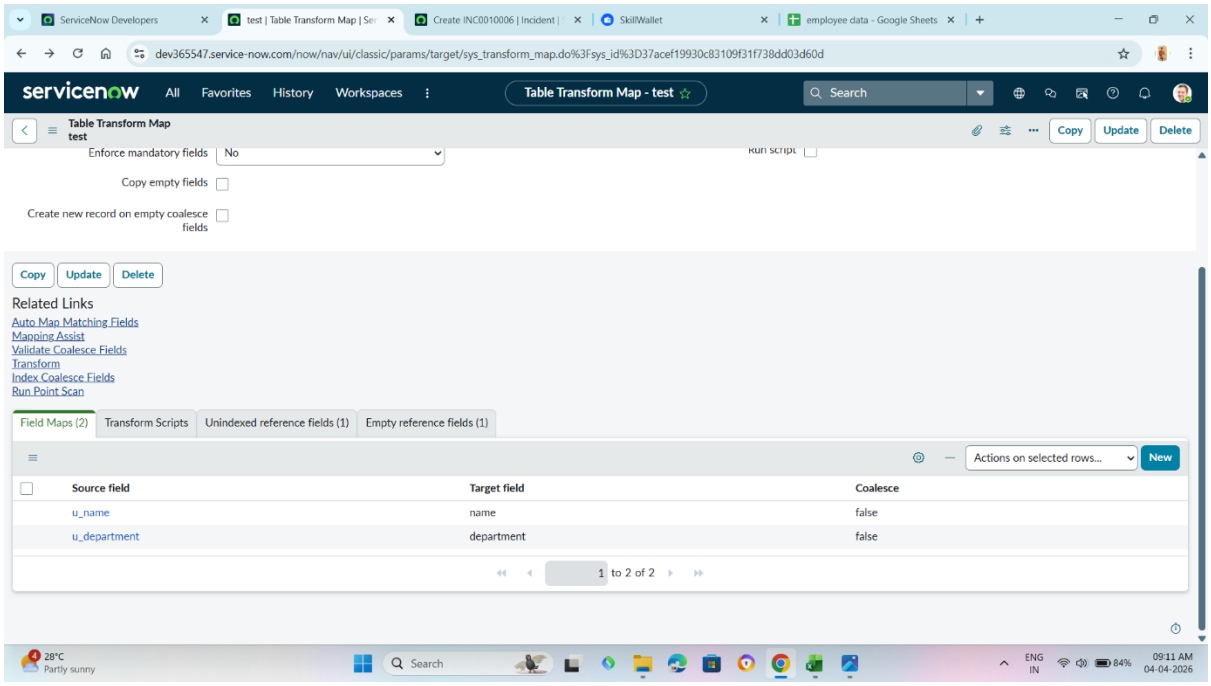
Match the fields using mapping assist:

Scrolled down the created transform map form

Click on auto matching fields

Then in the mapping fields some fields are matched





Using mapping assist map the manager field.

Select the fields which you are mapping in the staging table and target table

Select manager in source table and add

Select manager field in target table

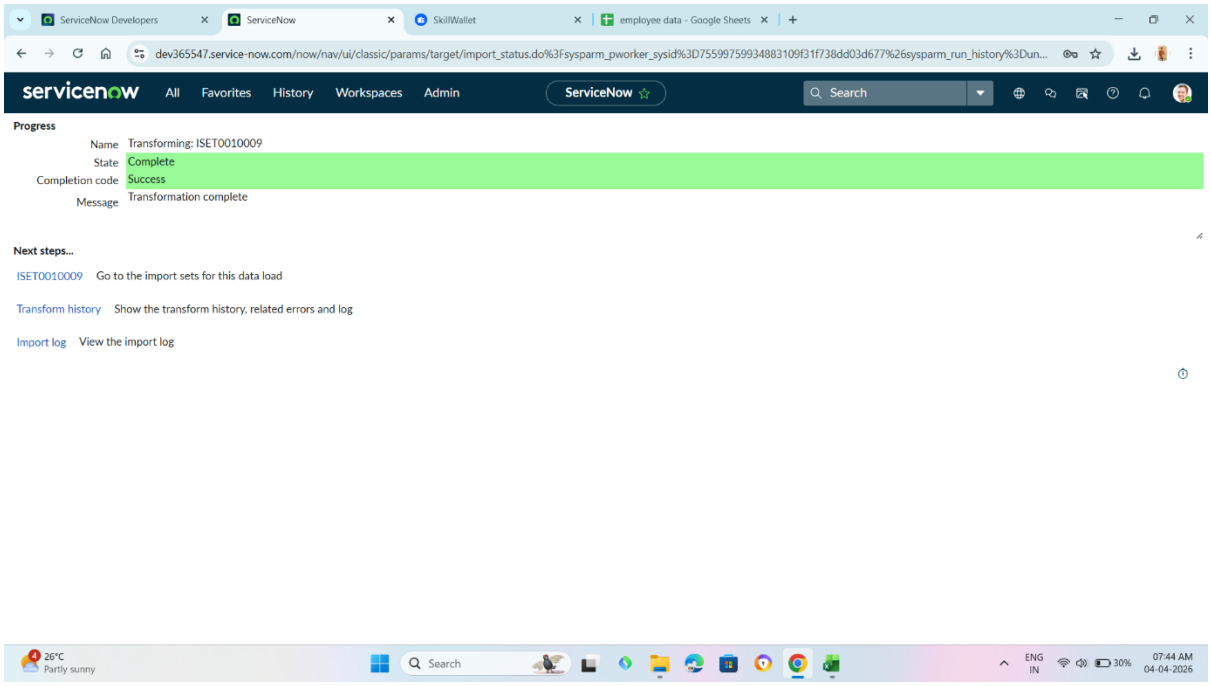
Click on the save button.

Transform the Data:

Click on transform in the related links.

Click on transform.

Data's transform state is completed.



Dot Walking:

Display department to the assigned to user

Open service now.

Click on All >> incident

Select create new under incidents

There is no field like department in the form

Email and manager fields are not visible

By using dot walking retrieve the data without opening the record

Click on context menu go to configure and select form layout

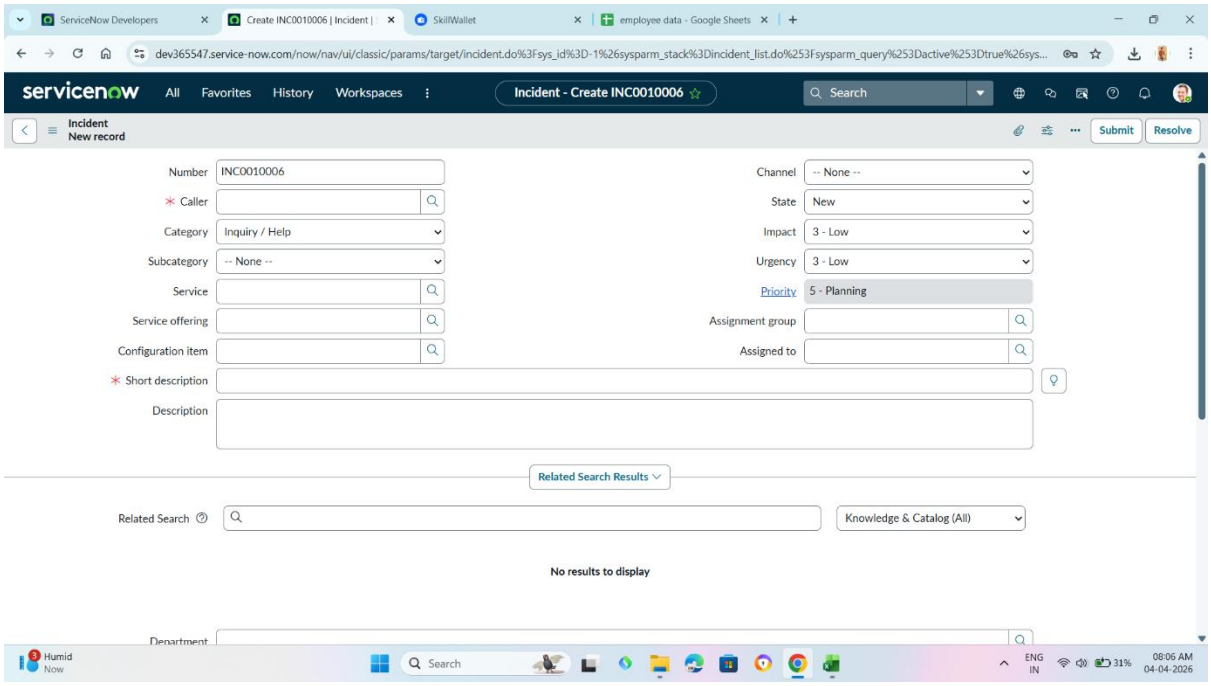
Select assigned to.department as mentioned below

Click on save

Add another field assignment.email.

Also add assignment.manager field

Click on save.

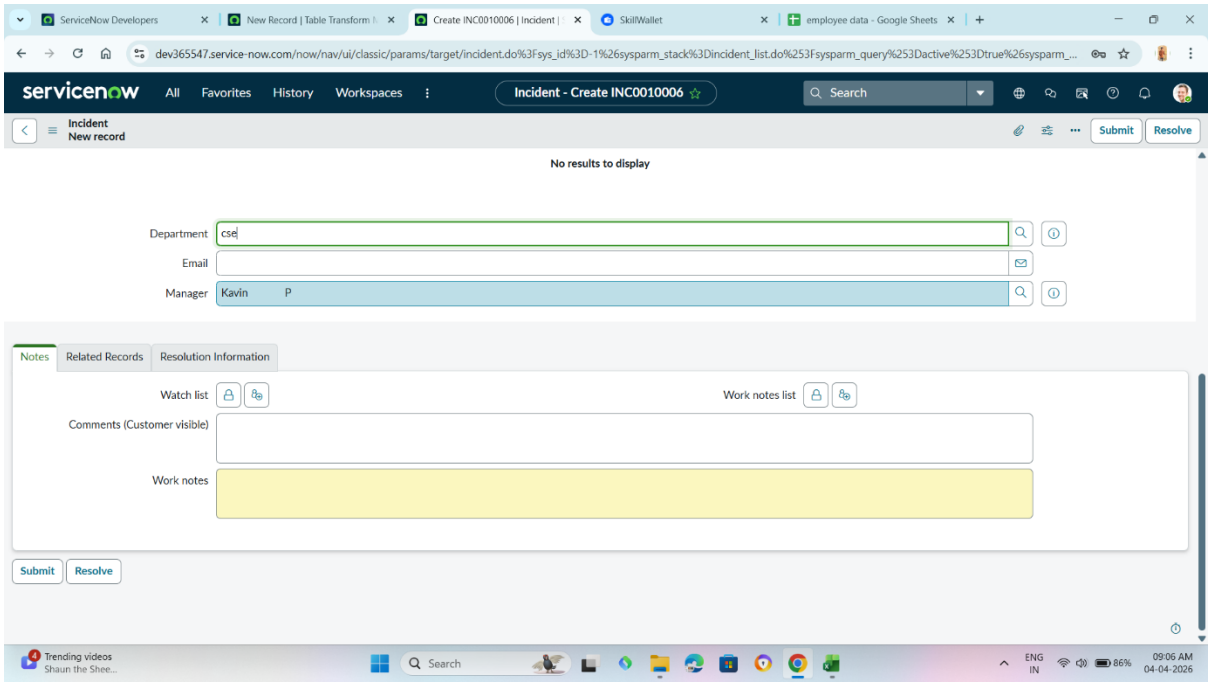


Open an incident form

14. Fill the mandatory fields and assigned to just imported user
15. Then department and email field is automatically filled

Testing:

Testing is the process of verifying that a system, feature, or functionality works as expected and meets defined requirements before it is released to users.



Conclusion:

Automating data import using **Import Sets and Transform Maps** ensures efficient and safe data handling, while **Dot Walking** enables seamless access and mapping of related records. Together, they provide a powerful mechanism to manage complex data relationships in enterprise systems.